

Cryptography: Post-Quantum versus Classical

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Overview

- Objective: Compare classical cryptography with post-quantum cryptography (PQC)
- Scope: Survey of major PQC techniques
- Motivation: Ensuring future-proof secure communication
- Keywords: Cryptography | Post-Quantum | Lattice-Based | Security | Communication

Quantum Threats

- Rapid advancement of quantum computing
- Shor's algorithm breaks RSA & ECC
- "Harvest-now, decrypt-later" attack model
- Implications for data confidentiality and integrity

Classical Cryptography – Core Concepts

- Definition: Secure communication in the presence of adversaries
- Encryption: Plaintext $M \rightarrow$ Ciphertext C
- Decryption: Ciphertext $C \rightarrow$ Plaintext M
- Correctness: $D(E(M))=M$
- Symmetric vs. Asymmetric key schemes

Classical Cryptography – Four Pillars of Security

Pillar	Definition	Mechanism
Confidentiality	Prevent unauthorized data access	Encryption algorithms
Integrity	Detect unauthorized data modification	MACs, cryptographic hashes
Authentication	Verify identities of communicating parties	Digital signatures, certificates
Non-repudiation	Prevent denial of message origin or receipt	Digital signatures

Classical Cryptography – Key Algorithms

- Symmetric
 - One-Time Pad: Information-theoretic secrecy
 - Shift Ciphers (Caesar cipher)
 - Block ciphers (e.g., AES)
- Asymmetric
 - RSA: Integer factorization basis
 - Diffie-Hellman key exchange
 - Elliptic-curve cryptography (ECC)

Why Post-Quantum Cryptography?

- Quantum computers can solve hard problems (factoring, discrete log)
- Current public-key systems (RSA, ECC) become insecure
- PQC aims for algorithms resilient against quantum attacks
- Proactive migration needed to protect “long-lived” secrets

PQC Algorithms

- Lattice-Based Cryptography
 - Security: Learning With Errors (LWE) hardness
 - Pros: High efficiency, straightforward implementation
 - Cons: Larger key sizes
- Code-Based Cryptography
 - Security: Decoding general linear codes
 - Example: Classic McEliece
 - Pros: Long-standing attack resistance
 - Cons: Very large public keys

PQC Algorithms

- Hash-Based Cryptography
 - Security: One-way hash functions
 - Example: SPHINCS+
 - Pros: Simple security proofs
 - Cons: Slower signing and verification
- Multivariate Polynomial Cryptography
 - Security: Solving systems of nonlinear equations
 - Examples: Rainbow, HFE
 - Pros: Short signatures
 - Cons: Some schemes have been broken
- Isogeny-based cryptography

Conclusion & Urgency

- Quantum computing progress threatens today's cryptosystems
- NIST leading standardization of PQC candidates
- Early adoption and hybrid approaches as transition strategy
- Ongoing research needed for performance and key-size optimizations

Future Work

- Further optimization of key and signature sizes is required to make PQC schemes practical for widespread use.
- Development of quantitative frameworks to balance security levels against computational and bandwidth costs will guide implementation decisions.
- Designing mixed quantum/classical authentication protocols can leverage the strengths of both paradigms.
- Institutions should formulate comprehensive migration roadmaps and update policies to ensure timely adoption of recommended standards.

Thank you